$$\nu(\tau) = \int_0^\infty \mathrm{d}\Delta\, J(\Delta) \cos\left(\Delta\tau\right), \tag{5.191}$$

$$\eta(\tau) = \int_0^\infty \mathrm{d}\Delta\, J(\Delta) \tanh\left(\frac{\Delta}{2k_\mathrm{B}T}\right) \sin\left(\Delta\tau\right), \tag{5.192}$$

obtained by inserting the new effective spectral density (5.189) into (5.16) and (5.17). The formal differences between these expressions for the kernels